Finance 6110: Corporate Finance
Final Exam

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True/False and Multiple Choice

1. FALSE	2. TRUE	3. TRUE	4. TRUE	5. TRUE
6. TRUE	7. FALSE	8. FALSE	9. TRUE	10. TRUE
11. C	12. B	13. D	14. A	15. B
16. C	17. D	18. C	19. A	20. B

Problem 1: Ditta Project Valuation

a. Cost of equity = 18%
b. WACC = 12.4%
c. See table below
d. -$86.3319 (see PV for each FCFF below)

t =	*0*	*1*	*2*
EBIT		$5	$8
D&A		$3	$3
CAPEX	$100		
FCFF(part c)	$(100.00)	$7.00	$9.40
PV(FCFF) (part d)	$(100.00)	$6.23	$7.44

e. TV = $92.1923, PV(TV) = $72.9730
f. NPV = -$13.3589, no NPV is negative, they should not proceed.
g. Yes, it doesn't hurt the firm to have the option, it can only help. The ability to cancel if it's not going well and continue if it is increases the project's value.

Problem 2: Lumon M&A

a. For PE: $PE_N = 15$, $PE_S = 18$ (avg = **16.5**). For PB: $PB_N = 5$, $PB_S = 5.4$ (avg = **5.2**)
b. Value under PE = $16.5 \times 4M = \$\mathbf{66M}$, value under PB = $5.2 \times 15M = \$\mathbf{78M}$
c. PE is a better multiple. Price to book is harder to compare across intangibles-heavy companies like those in the biotech industry, as intangibles are often a key source of value but tend to be undervalued in measures of book equity.
d. The premium is \$80M - \$66M = **\$14M** (if they used PB, they'll have \$2M)
e. The target gains **\$14M** of the synergies, the acquirer gains the remaining \$30M - \$14M = **\$16M** (if they used PB, they'll have \$2M and \$28M).
f. **No**, it's not unusual. As we discussed in class, typically the target's price rises after a deal is announced and the acquirer's price falls. This suggests investors suspect the acquirer is at risk of over-paying for the target.

Problem 3: Recapitalization at Globex

a. $EV = E + D - cash = 200M * \$15 + 0 - 0 = \$\mathbf{3B}$
b. $EV' = \$3B$ (no change under perfect capital markets). $E' = \$\mathbf{1.8B}$
c. $EV' = \$2.5B + \$1.2B = \$\mathbf{3.7B}$
d. $PV(TS) = D\tau = \$1.2B * 35\% = \$\mathbf{0.42B}$
e. $PV(other) = 0.7B - 0.42B = \$\mathbf{0.28B}$
f. Since firm value rose by more than expected from the tax shield benefits alone, there must be another trade-off that increases value when leverage rises. Among the options given, **money-to-burn** best explains this.
g. Money-to-burn occurs when an abundance of cash leads managers to make worse investment choices, such as pet projects or wasteful spending. Increasing leverage makes cash more scarce by effectively earmarking future cash to be paid out to debtholders, reducing the money-to-burn problem.
h. **Yes**, his logic is flawed. The cost of equity rises when leverage increases, even if capital markets are perfect.

Problem 4: Hedging Corn

a. Exact value = \$5.40. If they use $\approx S_0(1 + r + c)^T$, they'll get 5.39 (you can accept this answer)
b. Short corn futures, long corn puts
c. **For short corn futures:** y-intercept = futures price, downward sloping with slope = -1, x-intercept is futures price. **For long corn put:** y-intercept is strike price,

sloping down with a slope of -1, x-intercept is the strike price. To the right of the strike it's flat.

d. Long put and short call.
e. **$0.79**, they'll get $0.80 if they use their rounded answer from the approximation in a (that's ok).
f. **Yes**. Sell the portfolio (long put, long corn, short bond) and buy the call.
g. Must list two of the five: (1) illiquidity; (2) short-selling constraints; (3) transaction costs/frictions; (4) model risk; (5) agency costs/conflicts of interest